A gas-transport model predicts the *Arabidopsis* spaceflight transcriptome, and illumination separates six flight contrasts

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Abstract

Plants flown on the International Space Station are grown inside enclosures, and the atmosphere inside those enclosures is rarely measured. We coupled a validated lattice-Boltzmann model of leaf boundary-layer gas transport to a Farquhar–von Caemmerer–Berry model of C_3 photosynthesis and asked what an illuminated, sealed canister does to the CO_2 reaching Rubisco. A mass balance with an explicitly parameterized headspace predicts that photosynthesis draws a sealed Biological Research in Canisters (BRIC) enclosure toward its CO_2 compensation point within minutes, reducing 12 h carbon gain to 1 % of an unenclosed control; at a drawn-down 100 ppm, the Rubisco oxygenation fraction $\varphi = V_o/(V_o + V_c)$ rises from 18 % to 46 % at experimental growth irradiance (60 µmol m^{-2} s^{-1}) and from 28 % to 54 % at saturating light. Because that drawdown is an enclosure-scale mass balance, it occurs in the ground control as well as in flight under symmetric conditions. It therefore does not bias the flight-versus-ground contrast every such experiment reports — it changes what that contrast is *about*. We estimate the enclosure effect on metabolite pools to be 22 times the microgravity effect, which means the spaceflight response is being measured in plants their own hardware has already carbon-starved. Contrary to our starting hypothesis, the microgravity boundary-layer penalty does not amplify as CO_2 falls; it converges, leaving an assimilation deficit at the experiment's irradiance of under 3 % at leaf and rosette scale and up to 11 % in a dense canopy (5–9 % and 26–29 % at saturating light). We tested these predictions against OSD-522 (BRIC-LED), where the model called eight of nine transcriptome gene-set responses and four of seven in the proteome, and against four further OSDR studies, six flight-versus-ground contrasts in all. On the starvation-over-photosynthesis contrast illumination separated the six completely — every illuminated contrast positive, every dark one negative, although the model predicted no signal in the dark rather than an inverse — while the predicted gradient across enclosure types did

not hold. NASA's Open Science Data Repository contains no plant metabolome at all (none of its twelve metabolite-profiling studies is of a plant), so the third omic layer for pathway integration was formulated as a falsifiable hypothesis scaffold. We argue that enclosure atmosphere is a first-order, currently unmeasured covariate in plant spaceflight biology — not a bias to correct for, but a baseline state to measure — and set out the three-arm design and in-canister logging that would resolve it.

Keywords: spaceflight, microgravity, photorespiration, *Arabidopsis*, growth hardware, multi-omics

1 Introduction

Interpreting a plant spaceflight experiment means attributing a molecular difference to microgravity. That attribution rests on an assumption that is almost never tested: that the flight and ground arms differ in gravity and in nothing else that matters to metabolism.

Plants flown on the International Space Station are grown inside enclosures — sealed Petri Dish Fixation Units in BRIC canisters, Petri plates on the cabin wall in CARA, actively ventilated chambers in VEGGIE and the Advanced Plant Habitat. These enclosures differ by orders of magnitude in how freely they exchange gas with the cabin, and to our knowledge no BRIC-class flight has logged the CO_2 concentration its plants actually experienced.

The quantity that matters is not the cabin atmosphere but the CO_2 mole fraction reaching Rubisco, C_c, which sits at the end of a chain of conductances: from bulk air across the leaf boundary layer (g_{bl}), through the stomata (g_s), and through the mesophyll (g_m). Only the first of those is gravity-dependent, through buoyancy-driven convection. Rubisco partitions between carboxylation and oxygenation as $V_o/V_c = 2\Gamma^*/C_c$ [1], so anything that lowers C_c raises photorespiration.

Here we ask a question about the enclosure rather than about gravity, and we ask it of the model before we ask it of any data.

2 Results

2.1 An illuminated sealed canister is a CO_2-starvation chamber

Boundary-layer conductance was taken from LunarLeaf-CFD [2], a lattice-Boltzmann solver validated against four analytical and empirical gates. The enclosure dynamics come from zero-dimensional (0-D) mass-balance models parameterized by geometry (Table S1). A headspace mass balance (30 mL air over 3 cm^2 leaf; well-mixed assumption) draws a lit BRIC canister from 400 ppm to near zero in approximately seven minutes; dynamic photosynthesis-feedback puts 12 h carbon gain at 1 % of an unenclosed 1 g control, against 90 % for CARA plates sealed with micropore tape (permeance 0.006 mol m^{-2} s^{-1}) and 100 % when ventilated (Fig. 1). Sensitivity analysis

demonstrates that rapid starvation holds across a wide envelope of leaf area (1–10 cm^2) and headspace (15–60 mL; Table S5).

Because drawdown is an enclosure mass balance, under symmetric thermal and sealing conditions it occurs in the ground control as much as in flight, canceling out of the published contrast.

2.2 What that does to Rubisco, and what gravity does not do

Feeding g_{bl} into a Farquhar–von Caemmerer–Berry model [1] with in-vivo kinetics [3, 4] at experimental irradiance (60 µmol $m^{-2}\,s^{-1}$, OSD-522) moves the leaf along its CO_2 response curve: oxygenation fraction $\varphi = V_o/(V_o + V_c)$ rises from 18.0 % in ventilated hardware at ambient CO_2 to 46.2 % in a drawn-down canister (at an assumed 100 ppm; Table 1, Fig. 2a). At saturating light (1,000 µmol $m^{-2}\,s^{-1}$), φ rises from 28.5 % to 53.7 %.

Contrary to our initial hypothesis, CO_2 starvation does not amplify the microgravity boundary-layer penalty (Fig. 2b). As assimilation falls, flux across the boundary layer (A/g_{bl}) vanishes and the gravities converge. What survives is an assimilation deficit of < 3 % at leaf and rosette scale (up to 11 % in a dense canopy) at 60 µmol $m^{-2}\,s^{-1}$, and 5–9 % (26–29 % in canopy) at saturating light (Fig. 2c). The sealed-versus-ventilated hardware effect on metabolite pools is 22 times the flight-versus-ground effect (Fig. 2d).

2.3 A metabolite layer that does not exist

Pathway-level integration needs a metabolome. An exhaustive census of all 567 OSDR studies revealed that 12 are metabolite profiling, and all 12 are mouse, human, rat or microbial (Table S3). **No plant study in OSDR has a metabolome.**

We therefore predicted a 21-compound layer as a falsifiable hypothesis scaffold (Fig. 3, Table S4), propagating fluxes under quasi-steady state: $\log_2(P_{\text{flight}}/P_{\text{ground}}) = \log_2(J_{\text{flight}}/J_{\text{ground}})$.

2.4 The model called eight of nine transcriptome gene sets

OSD-522 [5] grew *Arabidopsis* in BRIC-LED hardware under 60 µmol $m^{-2}\,s^{-1}$ light (6 flight vs 6 ground shoot libraries). Differential expression with PyDESeq2 [6] yielded 1,795 DEGs at FDR < 0.05 from 19,325 tested (1,658 had NA adjusted p due to standard DESeq2 independent filtering on low base counts; Fig. 4). Deposited proteomics provided ratios for 5,160 proteins.

Scoring against KEGG gene sets (Fig. 5), predictions held for eight of nine transcriptome sets: photosystems fell (-0.238, $p = 3.8 \times 10^{-6}$; proteome -0.820, $p = 5.0 \times 10^{-4}$), Rubisco fell (-0.403, $p = 4.5 \times 10^{-3}$), carbon fixation and starch/sucrose fell, DIN carbon-starvation markers rose ($+0.685$, $p = 2.6 \times 10^{-3}$), and photorespiratory enzymes were not induced. In set-versus-set contrasts immune to global depression, photorespiratory enzymes sat above Rubisco in transcriptome ($+0.239$, $p = 6.8 \times 10^{-3}$) and proteome ($+0.285$, $p = 4.9 \times 10^{-2}$).

2.5 Pathway integration puts photorespiration in the bottom 8 %

In PaintOmics [7] (Fig. 6), 32 of 232 pathways reached uncorrected $p < 0.05$, led by ER protein processing (UPR, $p = 8.6 \times 10^{-9}$). Photosynthesis was enriched ($p = 0.008$), while Glyoxylate and dicarboxylate metabolism (photorespiration) ranked 214th of 232 ($p = 0.994$, bottom 8 %). An integration control excluding predicted metabolites confirmed that these pathway rankings and the UPR dominance are fully invariant to the metabolite layer (Table S6). Independent enrichment analysis outside PaintOmics confirmed identical significance for photosynthesis and Calvin cycle repression (Table S7). In the metabolite hub analysis, 7 of 15 compounds reached FDR < 0.05, all carbon-starvation or supply nodes.

2.6 Illumination separates six contrasts; the enclosure gradient does not

Across six flight-versus-ground contrasts from five studies [8–11] (Fig. 7, Table 2), illumination separated them completely: all three illuminated contrasts were positive on starvation over photosynthesis ($+0.41, +0.92, +2.58$) and all three dark ones negative ($-0.39, -1.95, -3.22$; one-sided descriptive Mann–Whitney $p = 0.05$). The predicted gradient across enclosure classes did not hold ($\rho = -0.50, p = 0.67$), indicating that illumination driving drawdown, rather than enclosure sealing alone, dominates.

3 Discussion

The strongest signal in OSD-522 is unfolded protein response (UPR) in the ER, which gas transport does not explain. However, in an illuminated enclosure the hardware sets an unmeasured, extreme carbon-starvation baseline. Hardware cancellation between flight and ground is a baseline hypothesis that holds when gas exchange and thermal regimes are symmetric; any flight-ground asymmetry in leakage or temperature would expose hardware effects in the contrast.

3.1 Limitations

The cross-study comparison is descriptive and subject to platform and tissue variation. FvCB parameters reflect literature values rather than flight fits. Headspace drawdown was evaluated with 0-D mass balances; while verified across broad parameter sweeps, in-situ sensor logging remains essential.

3.2 What would settle it

First, log CO_2 and O_2 inside flight canisters. Second, profile a targeted 20-compound metabolite panel on archived material. Third, implement a three-arm design (flight sealed, ground sealed, ground vented) to separate main effects of hardware and gravity (Fig. 8), or a complete 2×2 factorial design (adding flight vented) to evaluate gravity $\times$ enclosure interactions.

4 Methods

Gas transport. Conductances g_{bl} were obtained from LunarLeaf-CFD [2] steady-state exports across leaf, rosette, and canopy scales. Enclosure drawdown kinetics were calculated using 0-D mass balance models (Table S1). Upscaling from leaf to rosette accounts for 8 overlapping blades with crown sheltering; canopy scaling models dense upright shoots (3.5 mm spacing) where boundary layers merge in parallel.

Photosynthesis model. The FvCB C_3 model was solved for supply-equals-demand fixed points across C_a grids and irradiances (Table S2). Oxygenation fraction is defined as $\varphi = V_o/(V_o + V_c)$. Surface O_2 excess from CFD ($\Delta O_2 \sim 5$–12 ppm) was propagated into local Γ^*.

Differential expression. OSD-522 was processed with PyDESeq2 [6] using design `~ condition` and Cook's distance refitting. Transcripts with NA adjusted p-values reflect standard independent filtering on low base counts.

Gene sets and pathways. KEGG pathway membership was retrieved live [12, 13]. PaintOmics 4 [7] was executed on KEGG and MapMan databases.

4.1 Use of AI tools

Analysis scripts, figures, and manuscript drafts were developed with AI coding assistants (Claude Code, Anthropic; Google Antigravity) under the author's direction. The author verified all data, derivations, and references, and takes full responsibility.

Supplementary information. Supplementary Tables S1–S7 and Supplementary Figure S1 accompany this paper.

Data availability. Primary data are available in NASA OSDR [14] (OSD-522, OSD-38, OSD-321, OSD-678, OSD-427). Derived tables are in the repository.

Code availability. Analysis code is available at https://github.com/dr-richard-barker/Photorespiration_multiomics_microgravity.

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Author contributions. R.B. conceived the study, conducted modeling and multi-omics analysis, and wrote the manuscript.

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Competing interests. The author declares no competing financial or non-financial interests.

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Table 1 Modelled operating points at experimental (60 µmol m^{-2} s^{-1}) and saturating (1000 µmol m^{-2} s^{-1}) light. g_{bl} in mol m^{-2} s^{-1} (two-sided leaf basis); C_a, C_c in µmol mol^{-1}; A in µmol m^{-2} s^{-1}; $\varphi = V_o/(V_o + V_c)$ in %. Rosette scale.

Arm	Irradiance (Q)	g_{bl}	C_a	C_c	φ (%)	A
Ground control (sealed)	60	0.540	100	88.2	45.8	0.94
Flight (sealed)	60	0.272	100	87.2	46.2	0.88
Ventilated (ambient)	60	0.272	400	334.6	18.0	4.90
Ground control (sealed)	1000	0.540	100	75.5	53.1	2.40
Flight (sealed)	1000	0.272	100	73.6	53.7	2.20
Ventilated (ambient)	1000	0.272	400	215.0	28.5	15.40

Table 2 The five OSDR studies analysed (six flight-vs-ground contrasts), with complete experimental metadata and confound and the six excluded studies.

Study	Hardware	Class	Tissue	Genotype	Duration	Light	Status
OSD-522	BRIC-LED	sealed	shoots	Col-0 WT	11 d	lit (60 µmol)	Included
OSD-38	BRIC-PDFU	sealed	whole seedlings	Col-0 WT	8 d	dark	Included
OSD-321	BRIC-PDFU	sealed	seedlings	WT	14 d	dark	Included
OSD-678	CARA square dish	tape	leaves	Col-0 WT	12 d	lit (cabin)	Included
OSD-678	CARA square dish	tape	leaves	Col-0 WT	12 d	dark	Included
OSD-427	VEGGIE (VPS)	vented	leaf	WT	21 d	lit (120 µmol)	Included
OSD-120	CARA square dish	tape	root tips	Col-0	12 d	lit/dark	Excluded (roots)
OSD-193	APH/APEX	vented	roots	Col-0, Sku6	14 d	lit	Excluded (roots)
OSD-281	APH/APEX	vented	roots	WS, Sku5	14 d	lit	Excluded (roots)
OSD-218	APH/APEX	vented	unstated	Col-0, WS	14 d	lit	Excluded (tissue unstated
OSD-16	APEX-TAGES	vented	leaf/stem	WT	12 d	lit	Excluded (no processed fi
OSD-7	TAGES	vented	whole	WT	12 d	lit	Excluded (microarray)

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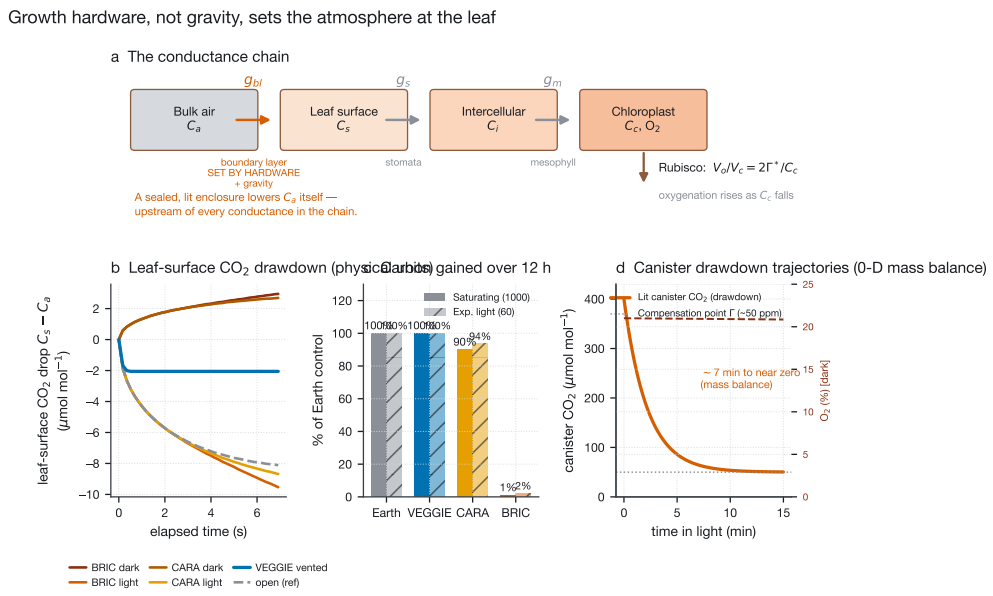


Fig. 1 Growth hardware, not gravity, sets the atmosphere at the leaf. **a**, The conductance chain. **b**, Simulated leaf-surface CO_2 difference ($C_s - C_a$, μmol mol^{-1}) over elapsed physical time (dt = 0.173 ms). **c**, Integrated 12 h carbon gain (% of Earth control) comparing saturating light (1000 µmol m^{-2} s^{-1}) and experimental light (60 µmol m^{-2} s^{-1}). **d**, Dynamic 0-D canister mass-balance trajectories showing rapid CO_2 drawdown to compensation point in lit canisters and gradual O_2 depletion in dark canisters.

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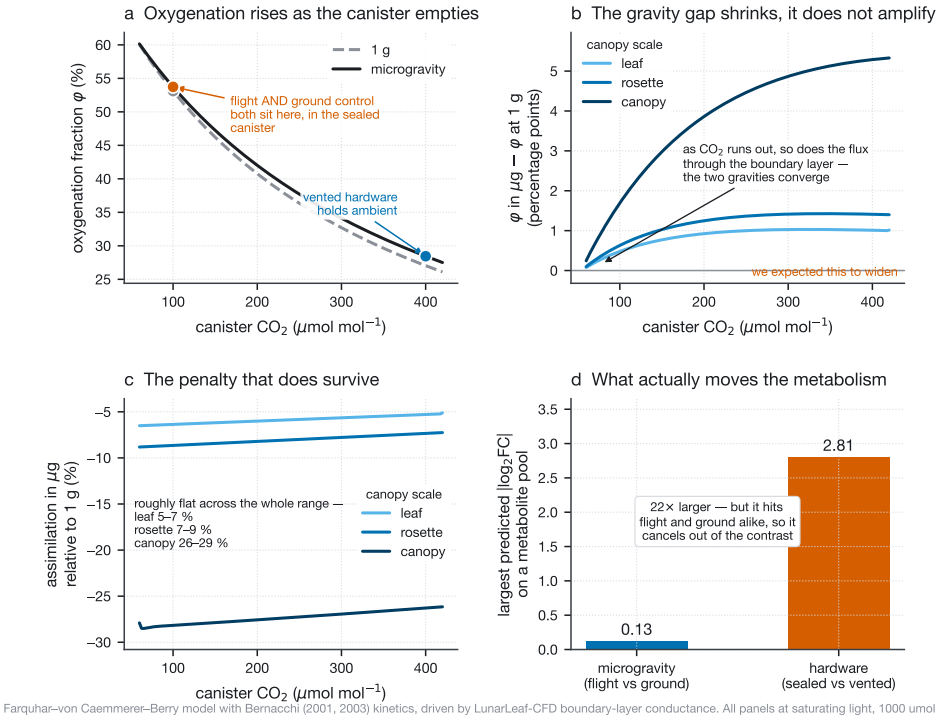


Fig. 2 A drawn-down enclosure, not microgravity, is what reaches Rubisco. **a**, Oxygenation fraction $\varphi = V_o/(V_o + V_c)$ at experimental light (60 μmol m^{-2} s^{-1}, solid) and saturating light (1000, dotted). **b**, The gravity gap in oxygenation shrinks toward zero as CO_2 falls. **c**, Proportional assimilation deficit ($< 3\%$ at rosette scale under 60 μmol m^{-2} s^{-1}; 7–9% at saturating light). **d**, The 22-fold dominance of the enclosure effect over the microgravity effect.

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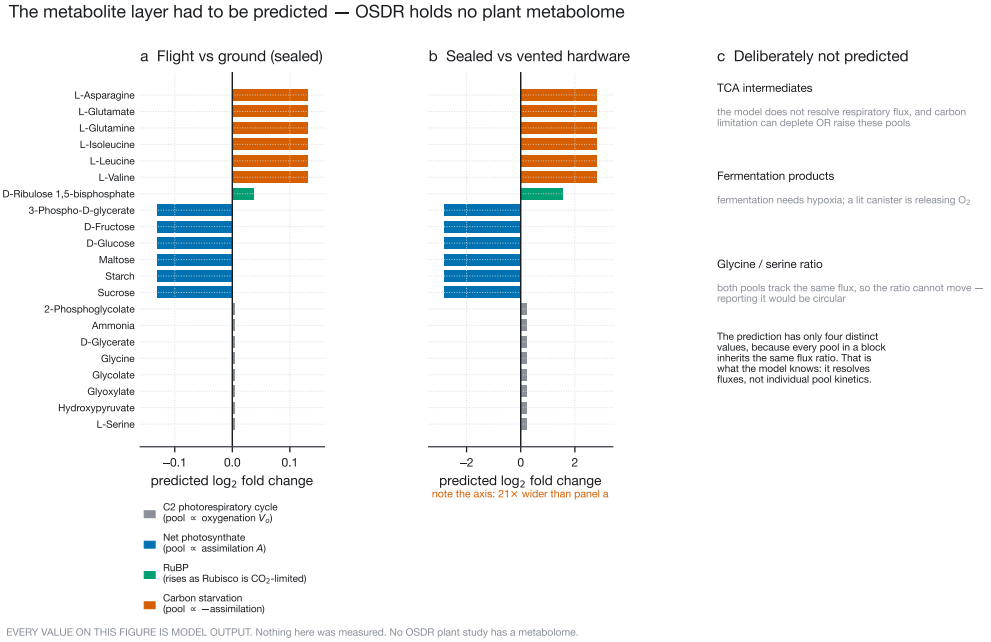


Fig. 3 The predicted metabolite layer. Formulated as a transparent hypothesis scaffold driven by modelled fluxes ($V_o, A, -C_c, -A$). Every value is model output.

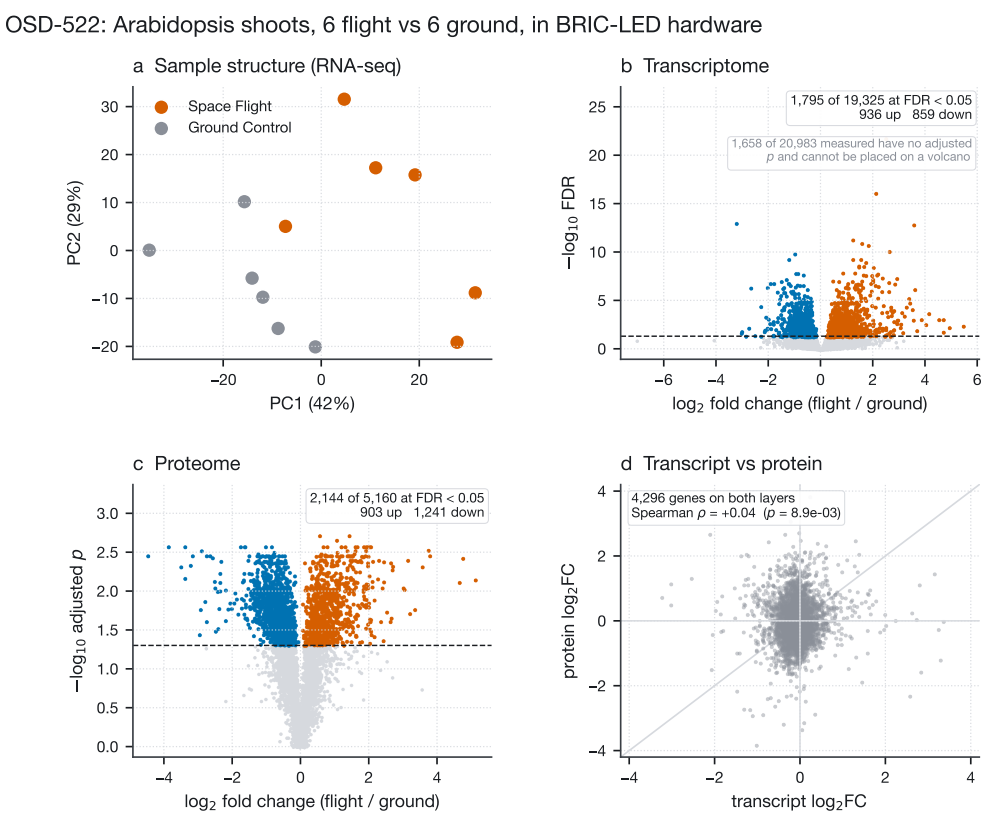


Fig. 4 OSD-522 omics profiling: 6 flight vs 6 ground shoot samples in BRIC-LED hardware. **a**, PCA of RNA-seq libraries. **b**, Transcriptome volcano (PyDESeq2; 1,658 transcripts with NA adjusted p due to independent filtering). **c**, Proteome volcano. **d**, Transcript vs protein correlation for 4,296 co-measured genes.

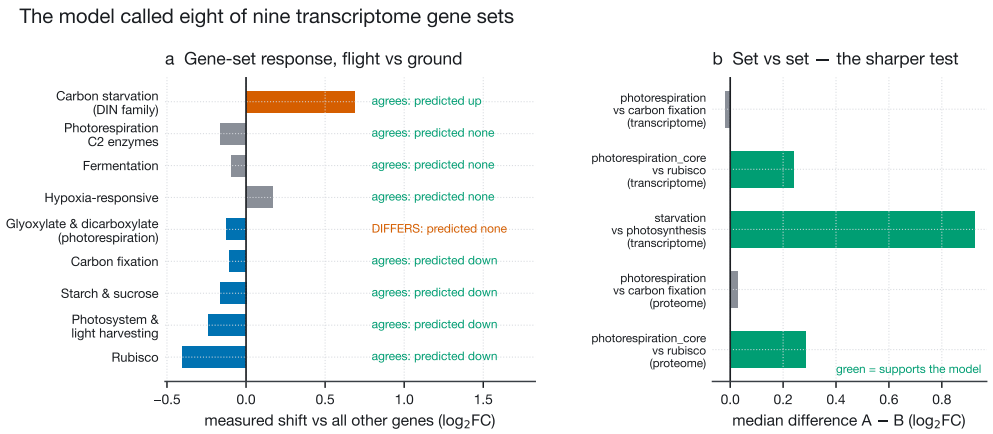


Fig. 5 The model's blind predictions against OSD-522 measurements. **a**, Directional calls across nine transcriptome sets. **b**, Set-versus-set contrasts demonstrating photorespiratory maintenance over photosynthetic apparatus.

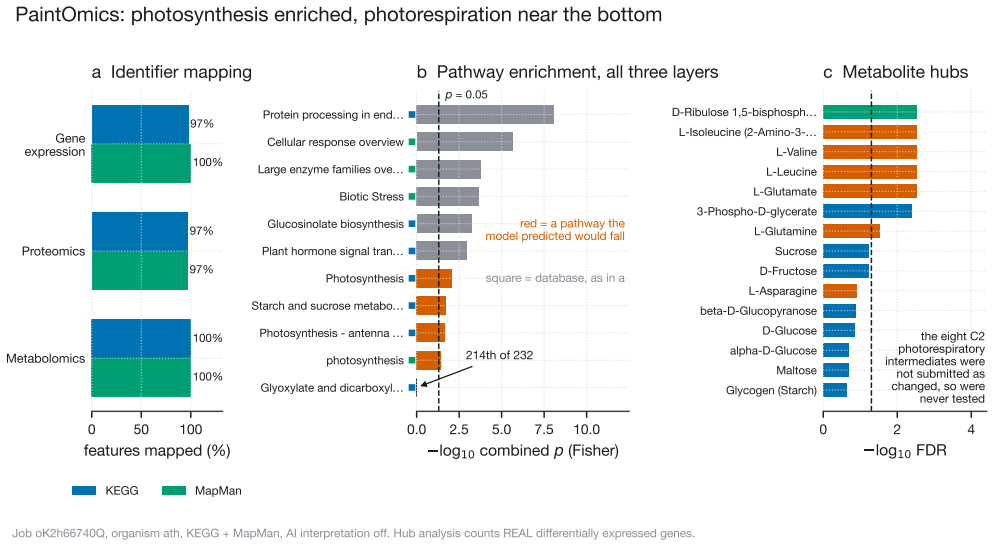


Fig. 6 PaintOmics multi-omics pathway integration. **a**, Feature mapping across databases. **b**, Pathway enrichment ranking: ER protein processing (UPR) dominates; photorespiration ranks in the bottom 8% (214th of 232). **c**, Metabolite hub analysis ranking predicted compounds by measured DEG neighbours.

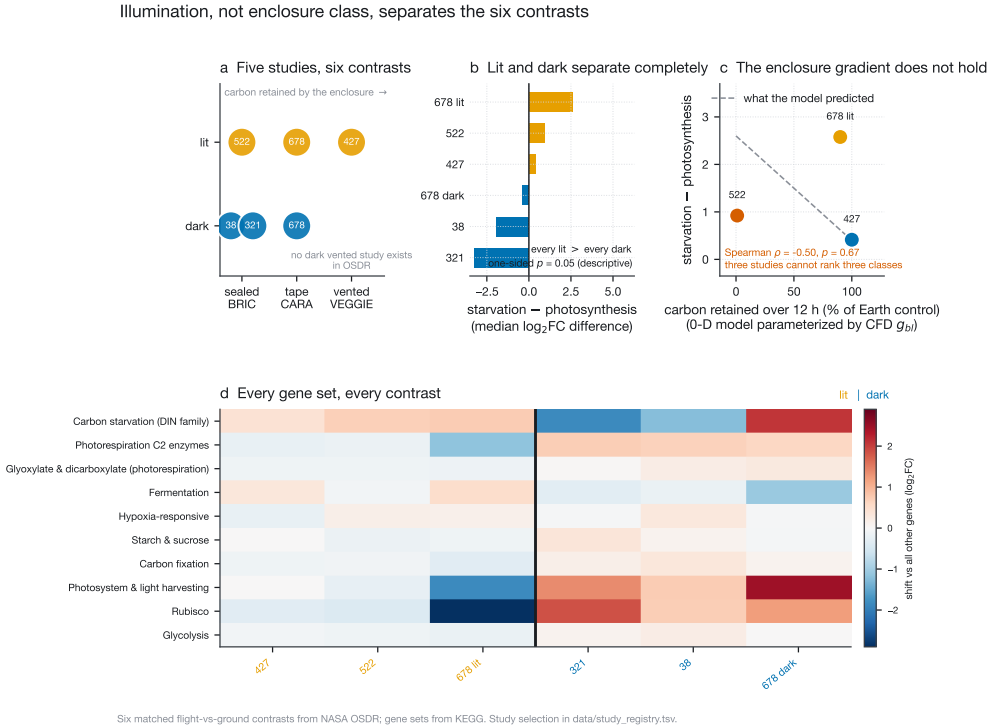


Fig. 7 Cross-study evaluation of the hardware ladder. **a**, Grid of six contrasts across five studies. **b**, Illumination separates all lit from all dark contrasts on starvation over photosynthesis. **c**, The predicted enclosure gradient does not hold (0-D model parameterized by CFD g_{bl}). **d**, Response matrix across all ten evaluated gene sets.

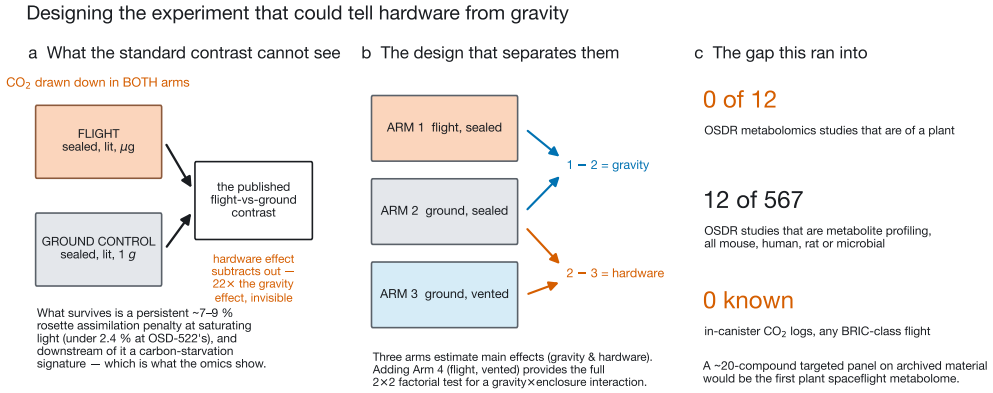


Fig. 8 Experimental designs to resolve growth hardware from microgravity. **a**, The published contrast cancels hardware effects. **b**, Proposed three-arm design separates main effects of hardware and microgravity; adding a flight-vented arm forms a complete 2×2 factorial test for interaction. **c**, The archive gap in OSDR plant metabolomics and in-canister logging.